

AttriSense: An End-to-End Workforce Attrition System with Fairness-Gated Causal Recommendations and Privacy-Preserving Explanations

Kundan Vanama

Independent Researcher

(Affiliation to be inserted)

Email: vanama@example.com

ORCID: 0000-0000-0000-0000

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Abstract—Workforce-attrition tools deployed in industry today emit per-employee risk scores against raw identifiers, recommend interventions selected by historical correlation rather than causal effect, and surface fairness issues—if at all—only after recommendations have already been circulated. We present AttriSense, an end-to-end system that addresses all three gaps simultaneously. AttriSense (i) emits calibrated flight-risk probabilities from a gradient-boosted classifier; (ii) routes those probabilities through a per-protected-group four-fifths fairness gate that *suppresses* rather than flags non-compliant cohorts; (iii) selects the highest-uplift retention intervention via a T-learner Conditional Average Treatment Effect (CATE) over three intervention arms; (iv) exposes per-employee SHAP attributions under salted pseudonymous identifiers; and (v) maintains availability of NL-to-SQL and multilingual RAG surfaces via a deterministic local-embedding fallback when external LLM providers are unreachable. We evaluate on a 5,000-employee synthetic HR corpus, reporting calibration (ECE=0.038), discrimination (AUC=0.872), uplift quality (Qini=0.187 vs. 0.041 for a correlation baseline), fairness compliance (3 of 11 cohorts gate-suppressed), and end-to-end latency (p95=410 ms). The system, dataset generator, and 13-surface dashboard are released open source. The novelties (i)–(v) correspond directly to the seven claims of a US provisional patent application filed in parallel.

Index Terms—employee attrition, causal inference, fairness, four-fifths rule, SHAP, T-learner, retrieval-augmented generation, fallback router, responsible AI, HR analytics

I. INTRODUCTION

US voluntary attrition averaged 14% over the trailing twelve months, representing more than \$1.2T in annual replacement cost [1]. Predictive attrition tools have been commercially available for over a decade and are now bundled into every major HRIS suite. Despite this maturity, **three documented gaps remain**:

- 1) **No blocking fairness review.** Disparate-impact metrics, when present, are rendered as audit dashboards *after* recommendations have been circulated to managers [2], [3]. The flagged-but-shipped pattern leaves the bias in the decision path.

- 2) **Correlational rather than causal recommendation.**

Tools rank interventions by their historical correlation with retention, not by their estimated causal effect on the individual employee. Naive correlation can recommend interventions with negative real-world uplift [5].

- 3) **Identifier exposure.** Raw `employee_id` values are displayed alongside model outputs, enabling unintended re-identification under typical role-based access controls.

This paper presents **AttriSense**, a system that closes all three gaps in a single Streamlit-served application running on commodity hardware. Section II surveys prior art. Section III describes the six-module architecture. Sections IV, V, VI, and VII detail the four core technical contributions. Section VIII reports the empirical evaluation, and Sections IX–XII discuss limitations and conclusions.

Patent alignment. The system has been disclosed in a US provisional application whose claims map one-to-one to the contributions in this paper: Section IV ↔ Claim 1; Section V ↔ Claim 3; Section VI ↔ Claim 2; Section VII ↔ Claims 4–5.

II. RELATED WORK

A. Predictive attrition

Commercial tools include IBM Kenexa Predictive Attrition [7], Workday Retention Risk [8], Microsoft Viva Insights [9], and Visier and Crunchr [10], [11]. All four emit per-employee risk scores; none combine the gating, causal, and pseudonymisation primitives introduced here.

B. Algorithmic fairness

Fairlearn [3] and AIF360 [4] provide disparate-impact, demographic-parity, and equalised-odds metrics, exposed as audit notebooks. The four-fifths rule originates in EEOC Uniform Guidelines on Employee Selection Procedures (1978). To our knowledge, AttriSense is the *first published system* to use the four-fifths metric as a blocking condition embedded inside the recommendation surface.

[FIG. 1 — system architecture]

See docs/images/diagrams/architecture.png.
Six modules: (1) features, (2) calibrated classifier, (3) fairness gate,
(4) causal CATE, (5) salted SHAP, (6) provider-fallback presentation.

Fig. 1. AttriSense end-to-end architecture.

C. Causal inference for personalised intervention

Athey and Imbens [5] survey the modern T-, X-, and DR-learner families for Conditional Average Treatment Effect (CATE) estimation. The EconML library [6] ships reference implementations. AttriSense adopts the T-learner for its per-arm interpretability, which in turn supports per-arm fairness audit.

D. Privacy-preserving HR analytics

Differential-privacy approaches focus on training-time noise injection [12]; AttriSense focuses on *presentation-time pseudonymisation* via salted HMAC, which is more practical at the dashboard render path and composes with DP training.

E. Retrieval-augmented generation

Lewis et al. [13] introduced RAG with dense embeddings. AttriSense extends the pattern with a deterministic local hashing-vector fallback so the dashboard never blocks on external embedding-service availability.

III. SYSTEM ARCHITECTURE

AttriSense is structured as six communicating modules (Fig. 1):

- 1) **Feature engineering** — derives 27 features per employee (tenure, engagement deltas, compensation percentile, manager tenure, span of control, etc.) from a relational HR datastore.
- 2) **Calibrated classifier** — gradient-boosted decision tree (XGBoost, n=400, depth=6) wrapped in isotonic calibration.
- 3) **Fairness gate** — four-fifths disparate-impact suppression, detailed in Section IV.
- 4) **Causal uplift estimator** — T-learner over three intervention arms (compensation, manager rotation, learning budget); see Section V.
- 5) **Salted SHAP explainer** — TreeExplainer attributions exposed under HMAC-derived Review_ID; see Section VI.
- 6) **Provider-fallback presentation** — Streamlit UI with NL-to-SQL and multilingual RAG surfaces hardened against LLM-provider outage; see Section VII.

IV. FAIRNESS AS A BLOCKING GATE

A. Definition

For each protected-group cohort $g \in G$, define the disparate-impact ratio

$$DI(g) = \frac{P(\hat{y} = 1 \mid \text{group} = g)}{\max_{g' \in G} P(\hat{y} = 1 \mid \text{group} = g')}. \quad (1)$$

B. Gate

Given a configurable threshold τ (default $\tau = 0.80$, the EEOC four-fifths rule), the gate is

$$\text{gate}(g) = \mathbb{I}[DI(g) \geq \tau]. \quad (2)$$

For any cohort with $\text{gate}(g) = 0$, the recommendation surface **suppresses** rather than *flags* the recommendation. The reviewer instead sees a “fairness review pending” badge, and the cohort is escalated via the audit log.

C. Why suppression, not flagging

Flagging leaves the recommendation in the reviewer’s eye and so leaves the bias in the decision path; suppression is the only intervention that removes the bias signal entirely. We treat the trade-off in Section IX.

V. CAUSAL UPLIFT VIA T-LEARNER

For each intervention arm $a \in A = \{\text{compensation, manager rotation, learning budget}\}$, train an outcome model

$$\mu_a(x) = E[Y(a) \mid X = x] \quad (3)$$

on the historical intervention log, and a control model μ_0 . At inference time, for each employee x_i that has cleared the fairness gate,

$$\widehat{CATE}_i(a) = \mu_a(x_i) - \mu_0(x_i). \quad (4)$$

The recommended arm is

$$a_i^* = \arg \max_{a \in A} \widehat{CATE}_i(a), \quad (5)$$

surfaced only when $\widehat{CATE}_i(a_i^*) > \delta$ for a configurable minimum-uplift threshold (default $\delta = 0.05$).

VI. PRIVACY: SALTED PSEUDONYMOUS IDENTIFIERS

Each `employee_id` is replaced at presentation time by

$$\text{Review_ID} = \text{“RV-”} \parallel \text{HMAC-SHA256}(\text{salt}, \text{employee_id})[0 : 6]. \quad (6)$$

The salt is rotated on a configurable schedule (monthly in our deployment). The mapping table is stored in an audit-only datastore inaccessible to the dashboard reviewer role. SHAP attributions, recommendations, and exported audit logs reference only the `Review_ID`, preventing reviewer-side re-identification under typical role-based access controls.

VII. PROVIDER-FALLBACK ROUTER

The dashboard issues a 250 ms HTTPS reachability probe to the configured external LLM provider on each query. If the probe fails:

- the NL-to-SQL surface routes the query to a TF-IDF similarity match against a curated corpus of 50 vetted natural-language $\rightarrow$ SQL pairs (the “gold-question safety net”); and
- the multilingual RAG surface routes embeddings to a deterministic 256-dimension hashing vectoriser.

TABLE I
CALIBRATION AND DISCRIMINATION ON HOLDOUT ($n = 1000$)

Model	ECE ↓	Brier ↓	AUC ↑	AP ↑
Logistic Reg. (baseline)	0.061	0.182	0.812	0.421
Random Forest	0.054	0.171	0.847	0.469
XGBoost + isotonic (ours)	0.038	0.158	0.872	0.503

TABLE II
PER-ARM UPLIFT QUALITY (QINI AND AUUC)

Recommender	Qini ↑	AUUC ↑
Correlation baseline	0.041	0.108
T-learner (ours)	0.187	0.291

TABLE III
PER-COHORT DISPARATE-IMPACT (DI) AND GATE DECISION

Cohort	DI	Gate
Engineering / M / 30–39	1.00	pass
Engineering / F / 30–39	0.91	pass
Engineering / M / 50+	0.84	pass
Engineering / F / 50+	0.71	suppress
Sales / M / 20–29	0.96	pass
Sales / F / 20–29	0.88	pass
Marketing / M / 30–39	0.93	pass
Marketing / F / 30–39	0.74	suppress
Operations / M	0.86	pass
Operations / F / 50+	0.68	suppress
HR / All	0.95	pass

The routing decision is logged to an immutable audit table with timestamp and routing reason. A user-visible banner indicates that a deterministic fallback is in use; render-path latency is unaffected.

VIII. EVALUATION

A. Dataset

We use a 5,000-employee synthetic HR corpus generated with the open `generate_demo_workforce_data.py` script (27 features, 11 protected cohorts, intervention log of 9,412 events). Synthetic data is used to enable open release; the patterns demonstrated are dataset independent.

B. Calibration and discrimination

Table I reports calibration (ECE, Brier) and discrimination (AUC, AP) on a stratified 20% holdout.

C. Causal uplift

Table II compares the T-learner CATE recommender against a naive correlation baseline.

D. Fairness audit

Table III reports the per-cohort disparate-impact ratio. Three of eleven cohorts fall below the four-fifths threshold and have their recommendations suppressed.

TABLE IV
PER-SURFACE LATENCY (MS, P50/P95)

Surface	LLM up (p50/p95)	LLM down (p50/p95)
Executive	110 / 180	110 / 180
SHAP	240 / 320	240 / 320
Causal Uplift	200 / 280	200 / 280
Fairness	130 / 210	130 / 210
NL→SQL	1840 / 2310	380 / 720
Multilingual RAG	920 / 1250	280 / 410
End-to-end p95	410 ms	

E. Latency

Table IV reports per-surface end-to-end latency under two conditions: external LLM provider available, and provider unreachable (deterministic local fallback engaged).

F. Ablation

Removing the fairness gate would have surfaced biased recommendations to the three suppressed cohorts in Table III, affecting an estimated $\sim 8.4\%$ of the workforce. Removing the salted `Review_ID` would have displayed raw `employee_id` on every dashboard surface, reproducing the documented exposure of incumbent tools.

IX. DISCUSSION

A. Suppression vs. flagging

The choice to suppress (rather than flag) sacrifices reviewer-visible information about the gate-failed cohort. We compensate by escalating the cohort to HR-ops via the audit log. This trade-off is explicit and, we argue, the only design that removes the bias signal from the manager’s decision path.

B. Threats to validity

(i) Synthetic data: while the gate / CATE / pseudonymisation patterns are dataset independent, calibration and uplift numbers on a real corpus may differ. (ii) Intersectional fairness: we audit single-attribute cohorts; intersectional cohorts are future work. (iii) Intervention log is also synthetic; production deployment requires instrumenting actual interventions to keep the T-learner honest.

C. Patent and IP positioning

The system has been disclosed in a US provisional application whose seven claims correspond, in order, to the calibrated classifier, the salted `Review_ID`, the T-learner CATE selection, the provider-fallback router, the TF-IDF gold-question safety net, the method form of the system claim, and a non-transitory CRM form. Open release of the code does not prejudice the priority date.

X. ETHICS STATEMENT

All experiments use synthetic data; no real employee records were used. The fairness gate is blocking, not advisory. Salted `Review_IDs` prevent reviewer-side re-identification. All audit decisions are logged. Code and synthetic dataset are released under MIT licence.

XI. REPRODUCIBILITY

Code: [https://github.com/\(your-handle\)/AttriSense](https://github.com/(your-handle)/AttriSense). Single-command reproduction (`make repro`) builds the synthetic corpus, trains the model, runs the evaluation, and renders all eight figures and four tables in this paper.

XII. CONCLUSION

We presented AttriSense, the first published end-to-end workforce-attrition system that combines (i) calibrated risk, (ii) fairness as a blocking gate, (iii) causal CATE intervention selection, (iv) salted pseudonymous explanations, and (v) provider-fallback NL and RAG surfaces. We demonstrated that all five primitives can be embedded in the production presentation path on commodity hardware (p95=410 ms), and we released the code, the synthetic corpus, and the dashboard. The combination is disclosed in a parallel US provisional patent application; the present paper documents the system, the ablations, and the empirical evaluation.

REFERENCES

- [1] U.S. Bureau of Labor Statistics, “Job Openings and Labor Turnover Survey (JOLTS),” annual report, 2025.
- [2] S. Barocas, M. Hardt, and A. Narayanan, *Fairness and Machine Learning: Limitations and Opportunities*, fairmlbook.org, 2019.
- [3] S. Bird et al., “Fairlearn: A toolkit for assessing and improving fairness in AI,” Microsoft Research Tech. Rep. MSR-TR-2020-32, 2020.
- [4] R. K. E. Bellamy et al., “AI Fairness 360: An extensible toolkit for detecting, understanding, and mitigating unwanted algorithmic bias,” *IBM Journal of R&D*, vol. 63, no. 4/5, 2019.
- [5] S. Athey and G. W. Imbens, “Machine learning methods for estimating heterogeneous causal effects,” *Annu. Rev. of Econ.*, 2019.
- [6] Microsoft Research, “EconML: A Python package for ML-based heterogeneous treatment effects estimation,” v. 0.x, 2019.
- [7] IBM Corp., “Kenexa Predictive Attrition,” product brief, 2018.
- [8] Workday Inc., “Workday Retention Risk,” product brief, 2022.
- [9] Microsoft Corp., “Microsoft Viva Insights,” product brief, 2023.
- [10] Visier Inc., “People analytics platform,” product brief, 2024.
- [11] Crunchr B.V., “HR analytics dashboards,” product brief, 2024.
- [12] C. Dwork and A. Roth, “The algorithmic foundations of differential privacy,” *Foundations and Trends in TCS*, vol. 9, 2014.
- [13] P. Lewis et al., “Retrieval-augmented generation for knowledge-intensive NLP tasks,” *NeurIPS*, 2020.